

The World of Microorganisms

Tiny living beings, too small for the eye, found in water, soil, air — and even inside us.

What are microorganisms?

Microorganisms (microbes) are living beings too small to see with the naked eye — you need a microscope. *Micro* = very small; *organism* = living being. They live in water, soil, air, extreme hot springs and snow, and even inside our gut.

The main groups

Group	Key features	Example
Bacteria	Single cell; spherical, rod, comma or spiral shaped; have a nucleoid	Lactobacillus, Rhizobium
Protozoa	Single cell; often move with special structures	Amoeba, Paramecium
Fungi	Unicellular or multicellular; have cell wall but no chlorophyll	Yeast (uni), Bread mould (multi)
Algae	Contain chlorophyll (green); make their own food	Spirulina, Chlorella

Viruses are special. They are microscopic and **acellular** (not made of cells). They multiply **only inside a living host cell** and can cause disease. So they are *different from other microorganisms*.

Activities to remember

- Pond/stagnant water (Activity 2.4) shows protozoa & algae.
- Soil suspension (Activity 2.5) shows fungi, bacteria & algae.

Magnification: a microscope makes microbes look 100–400 times bigger so we can see them.

Where students lose marks

Trap 1 — Viruses are not "normal" microbes. They are acellular and reproduce only inside a host. Don't call them ordinary living cells.

Trap 2 — Yeast is a fungus, not a bacterium. Yeast (unicellular fungus) and mould (multicellular fungus) are both fungi.

Trap 3 — Algae are green; fungi are not. Algae have chlorophyll and make food; fungi lack chlorophyll.

Unicellular vs multicellular: bacteria & protozoa are unicellular; fungi can be either; algae one or more cells.

Model answers — write them like this

Example A. Why are viruses placed apart from other microorganisms?

Step 1: They are acellular — not made of cells.

Step 2: They reproduce only inside a living host cell.

∴ Being acellular and host-dependent makes viruses different from other microbes.

Example B. A green, single-celled organism that makes its own food — which group?

Step 1: "Green" → contains chlorophyll.

Step 2: Makes own food by photosynthesis.

∴ It is an alga (algae).

[Ready to practise?](#) → [Open the worksheet](#)